

This assignment will not be graded and is only for practice.

Level 1:

1) Consider a system of linear equations as follows:

1 point

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

Let the matrix representation of the above system be  $Ax = b$ , where  $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ ,  $x =$

$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ , and  $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ . Let  $A_{x_i}$  be the matrix obtained by replacing the  $i$ -th column of  $A$  (i.e.,  $\begin{bmatrix} a_{1i} \\ a_{2i} \\ a_{3i} \end{bmatrix}$ ) by

$b$ , for  $i = 1, 2, 3$ . If  $\det(A_{x_i}) = 0$  for  $i = 1, 2, 3$ , then which of the following is (are) true?

☐  $x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$  must be a solution.

☐  $x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$  is a solution if  $\det(A) = 0$ .

☐ If  $\det(A) = 0$ , then we cannot conclude about the solution using Cramer's rule.

☐ If  $\det(A) = 0$ , then  $b = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$  is a solution if  $\det(A) = 0$ .

If  $\det(A) = 0$ , then we cannot conclude about the solution using Cramer's rule.

If  $\det(A) = 0$ , then  $b = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ .

Consider a system of linear equations

$$\begin{aligned}x_1 + x_3 &= 1 \\ -x_1 + x_2 - x_3 &= 1 \\ -x_2 + x_3 &= 1\end{aligned}$$

Let matrix representation of the above system be  $Ax = b$ , where  $A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$ ,  $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ , and  $b = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ . Let  $A_{x_i}$  be the matrix obtained by replacing the  $i$ -th column of  $A$  (i.e.,  $\begin{bmatrix} a_{1i} \\ a_{2i} \\ a_{3i} \end{bmatrix}$ ) by  $b$ , for  $i = 1, 2, 3$ .

(Use the above information to answer questions 2 and 3)

2) Choose the set of correct options

**1 point**

☐  $A_{x_1} = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$

☐  $A_{x_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$

☐  $A_{x_2} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$

☐  $A_{x_3} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$A_{x_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$

$$A_{x_3} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$$

3) Choose the set of correct options.

**1 point**

- ☐  $x_1 = -2$ .
- ☐  $x_2 = -2$ .
- ☐  $x_3 = 3$ .
- ☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$x_1 = -2.$$

$$x_3 = 3.$$

## Level 2:

- 4) Consider the system of linear equations  $Ax = b$ , where  $A = \begin{bmatrix} 1 & a & 0 \\ a & 1 & 2 \\ 1 & 0 & a \end{bmatrix}$ ,  $x = \begin{bmatrix} x_1 \\ 0 \\ x_3 \end{bmatrix}$ , and  $b = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ ,

and the solution for  $x$  is partially known. What is the value of  $a^2$ , if  $a = 0, \sqrt{3}, -\sqrt{3}$  is given? [Hint: Observe that the second row of the vector  $x$  is given as 0, which implies that  $x_2$  is known, where  $x =$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

**1 point**

5) Consider the system of linear equations  $Ax = b$ , where  $A = \begin{bmatrix} 1 & 2a+3 \\ 3a+2 & 1 \end{bmatrix}$ ,  $x = \begin{bmatrix} x_1 \\ 0 \end{bmatrix}$ , and  $b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ , and the solution for  $x$  is partially known.

What is the value of  $3a$ , if  $\det(A) = 0$  is given?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -1

**1 point**

Your score is: 0/5